

EXP NO: 2	SUPPORT VECTOR MACHINE (SVM) AND RANDOM FOREST FOR BINARY & MULTICLASS CLASSIFICATION
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AIM

To build classification models using **Support Vector Machine (SVM)** and **Random Forest**, apply them to a dataset, and evaluate the models using **accuracy** and **confusion matrix** for binary and multiclass classification.

ALGORITHM

Part A: SVM Model

1. Import the necessary libraries.
2. Load and explore the dataset.
3. Handle missing values, if any.
4. Encode categorical variables.
5. Split the dataset into training and testing sets.
6. Build an SVM classifier using **SVC()**.
7. Train the model and make predictions.
8. Evaluate the model using accuracy and confusion matrix.

Part B: Random Forest Model

9. Initialize the Random Forest classifier using **RandomForestClassifier()**.
10. Train the model and make predictions.
11. Evaluate the model using accuracy and confusion matrix.
12. Compare the performance of SVM and Random Forest.

CODE:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_breast_cancer()

X = data.data
y = data.target

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=42
)

svm_model = SVC(kernel='linear')
svm_model.fit(X_train, y_train)

y_pred_svm = svm_model.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("\nSVM Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))

rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)

y_pred_rf = rf_model.predict(X_test)

print("\nRandom Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
```

```
print("\nRandom Forest Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))

plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.heatmap(confusion_matrix(y_test, y_pred_svm),
            annot=True,
            fmt='d',
            cmap='Blues')
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.subplot(1,2,2)
sns.heatmap(confusion_matrix(y_test, y_pred_rf),
            annot=True,
            fmt='d',
            cmap='Greens')
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.tight_layout()
plt.show()
```

OUTPUT:

SVM Accuracy: 0.9777777777777777

SVM Confusion Matrix:

```
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

Random Forest Accuracy: 1.0

Random Forest Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

